

Harsh Ranjan

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EDUCATION

University of California, Santa Cruz <i>B.S. Computer Engineering</i>	2022 – Dec 2026 Santa Cruz, CA
• Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Machine Learning, Statistics	
University of California, San Diego – Summer Session <i>Computer Architecture (CSE 142/L), Logic Design</i>	Summer 2025 San Diego, CA

EXPERIENCE

UC Santa Cruz – Baskin School of Engineering <i>Student Researcher – EADS (Electrical Anomaly Detection System)</i>	Sep 2026 – Present Santa Cruz, CA
• Supporting the EADS team under Dr. Pat Mantey , deploying AnyLog / EdgeLake nodes for distributed sensor data .	
UC Santa Cruz – Applied Mathematics & Statistics <i>Reader – Linear Algebra, Differential Equations, Statistics</i>	Fall 2025 – Present Santa Cruz, CA
• Graded coursework, wrote grading rubrics , and held weekly office hours across three math and statistics courses .	
Tesphase <i>Chief Technology Officer</i>	Apr 2025 – Sep 2025 San Francisco, CA
• Winner, SC Launchpad 2025 . Owned the roadmap for a React Native / Node.js / Supabase mobile app.	
• Ran 30+ customer interviews , then shipped authentication , real-time database sync , and push notifications end to end.	
Balat Enterprises <i>Product Management Intern</i>	Jun 2024 – Sep 2024 Chennai, India
• Turned customer feedback into product specs for a React / Firebase app, cutting post-release bugs 40% .	
• Redesigned an internal customer-service tool , cutting response times and lifting user retention 30% .	
Vyavshay Technologies <i>Marketing Intern</i>	Jul 2022 – Aug 2022 Chennai, India
• Analyzed campaign performance metrics to guide feature prioritization, contributing to a 15% lift in user retention .	

PROJECTS

CPU Performance Analysis & GPT-2 Optimization	C, Linux perf, AVX2, OpenMP
• Profiled GPT-2 training in C with Linux perf , cutting L1 cache misses 9% to ~2% via tiled matmul and AVX2 SIMD .	
• Measured multithreaded scaling with OpenMP and how cache and memory layout affect throughput.	
Microgrid Manager – Energy Disaggregation (Capstone)	Ongoing Python, PyTorch, Kafka, MQTT, PostgreSQL
• Streamed one-second sensor readings over MQTT into Kafka and PostgreSQL for live time-series monitoring.	
• Trained a BiLSTM model in PyTorch to disaggregate energy use across three sub-panels : HVAC, water heater, kitchen.	
• Built a rules-based labeling engine after LLM-based labeling failed, and stabilized ingestion on EdgeLake / AnyLog .	
• Writing a paper and presentation on the system for the EdgeLake Foundation .	
High-Performance HTTPS Proxy Server	Ongoing C, C++, Verilog
• Built a multi-threaded HTTP/HTTPS proxy in C/C++ using TCP/IP sockets , POSIX threads , and SSL/TLS .	
• Added FIFO and LRU caching with thread-safe memory management , cutting response time 50% for cached requests.	
• Porting the server onto FPGA hardware , bringing it up on Spartan-3E , Basys 3 , and Alchitry Io boards.	
Osmosis – VGA Molecule Game	FPGA, Verilog
• Designed a 640x480 60 Hz VGA controller and pixel renderer in Verilog with collision physics in RTL .	
• Randomized motion with an LFSR plus a player-set timer, with win/loss states driven by an FSM .	

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL, Verilog

Frameworks & Tools: PyTorch, React, React Native, Node.js, REST APIs, PostgreSQL, Kafka, MQTT, Docker, Git, CI/CD

Hardware: FPGA, RTL Design, Vivado, Basys 3, Spartan-3E, UART / SPI / I2C

Focus Areas: Distributed Systems, Performance Optimization, Machine Learning, Data Pipelines, Embedded Systems